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Incidence and risk factors for reproductive disorders in organic certified dairies

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ABSTRACT

The objective was to determine the lactational incidence risks of reproductive disorders (RD) including dystocia (DYS), retained fetal membranes (RFM), puerperal metritis (PM), clinical endometritis (CE), pyometra (PYO), late pregnancy loss (LPL), and culling due to reproductive reasons (CRR) in large organic certified dairies with predominant use of Holstein genetics. In addition, potential risk factors for RD, including parity category, milk yield in the previous lactation, season of dry-off, access to grazing at dry-off, length of the dry period (DPL), length of the close-up period (CUPL), calculated gestation length (CGL), season of calving, access to grazing at calving, concurrent health conditions, and total number of artificial inseminations were analyzed. A retrospective study was performed using information from 8 organic certified herds located in northern Colorado and west Texas, ranging from 900 to 6,000 milking cows. Cow information consisted of finalized lactations starting with calvings occurring between January 1, 2016, and December 31, 2018. Data were analyzed using time-to-event analyses and multivariable logistic regression. Clinical endometritis had the greatest lactational incidence risk (incidence risk = 10.1%; 95% CI = 9.8–10.3%), followed by DYS (8.8%; 8.5–9.1%), PM (4.9%; 4.7–5.1%), PYO (4.0%; 3.8–4.1%), LPL (3.6%; 3.4–3.7%), and RFM (2.0%; 1.8–2.1%). The incidence risk of CRR was 7.5% (7.3–8.7%), representing 29.1% (28.4–30.0%) of all culling reasons. Median time from calving to diagnosis of PM, CE, PYO, and LPL were 6, 23, 111, and 228 d, respectively. Median time for CRR was 391 d. The main risk factors for RD were DPL, CGL, and CUPL. Season of dry-off was a risk factor for RFM, PM, CE, LPL, and PYO. Short DPL (<30 d) was associated with increased odds (95% CI) of RFM [2.1 (1.6–2.7)], PM [1.4 (1.2–1.7)], and LPL [2.5 (1.5–4.2)]. Shortened CGL (<270 d) increased the odds of DYS [1.5 (1.3–1.8)], RFM [3.2 (2.4–4.1)], and PM [2.4 (1.7–3.3)], whereas extended CGL (>285 d) increased the odds of DYS [1.5 (1.4–1.7)], PM [1.5 (1.2–2.0)], CE [1.5 (1.1–2.0)], and CRR [1.2 (1.10–1.4)]. Shortened CUPL was associated with increased odds of all the RD. Cows spending less than 7 d in the close-up groups had greater odds of DYS [1.4 (1.1–1.8)], RFM [2.1 (1.8–2.4)], PM [1.9 (1.6–2.1)], PYO [1.5 (1.2–1.8)], LPL [2 (1.7–2.3)], and CRR [1.4 (1.04–1.8)] compared with cows spending more than 14 d in the close-up group. Season of calving was associated with the odds of PM [summer = 1.13 (1.01–1.2)], PYO [summer = 1.7 (1.1–2.5)], and LPL ([fall = 0.8 (0.6–0.96)] compared with winter calving. Other risk factors affecting RD were parity number, total number of artificial inseminations, and concurrent health conditions. Overall, lactational incidence risks of reproductive diseases were comparable with those reported in conventional herds. The identified risk factors for RD open opportunities for interventions, especially focused in precalving management, estimation of due dates, and closer monitoring of pregnancies from natural services in large organic certified dairy herds.
Key words: organic, reproductive disease, risk factors

INTRODUCTION

The demand for organically produced food has increased during the last decade in the United States. As a result of the growing demand for organic dairy products, the number of organic certified dairies (**ORG**) in the United States has increased to more than 2,500 farms with about 280,000 organic cows (McBride and Greene, 2009; Ruegg, 2009; USDA, 2017). In the United States, organic agriculture is regulated through the National Organic Program (USDA-NOP, 2016), which defines the rules under which farmers produce organic products or livestock. For ORG, regulations consist of common restrictions and requirements, regardless of the herd size, location, and management practices (NOP, 2019).

Many aspects of dairy production under organic certification are unique and result in specific challenges. Among other requirements, ORG must provide grazing

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for at least 120 d/yr, with at least 30% of the dry matter intake coming from pastures. In addition, outdoor access in yards or feeding lots is required, and the use of multiple drugs, including hormones (excluding oxytocin for postpartum disorders), antibiotics, and growth promoters is strictly prohibited (NOP, 2019). Although general challenges in reproduction may be similar for conventional and organic systems, the singular restrictions and regulations for organic certification result in different management scenarios (Pinedo and Velez, 2019).

As reproductive disorders (**RD**) result in impaired health and welfare, reduced fertility, and increased risk of culling (LeBlanc et al., 2002; Sheldon et al., 2009; Stojko et al., 2015; Gilbert, 2016), an adequate understanding of the dynamics of RD is critical for ORG production. However, research focused on reproductive health and performance in these systems in the United States and worldwide remains scarce (Fall et al., 2008). Moreover, as organic dairies present some distinctive features, such as lower milk yield per cow (Zwald et al., 2004; Nauta et al., 2006; Pol and Ruegg, 2007), greater use of crossbred cows (Buckley et al., 2014), and smaller herd sizes, comparisons in reproduction between organic and conventional farms are complex (McBride and Greene, 2009).

Most available studies comparing ORG and conventional operations refer to European operations and, despite showing disparity on their conclusions, they generally indicate lower reproductive performance and greater longevity in ORG compared with conventional dairies (Richert et al., 2013; Stiglbauer et al., 2013; Mullen et al., 2015; Sorge et al., 2016). Nonetheless, an extrapolation of these findings to ORG in the United States seems impractical due to significant differences in multiple features among these operations (Reksen et al., 1999; Hardeng and Edge, 2001; Fall et al., 2008). Although 45% of ORG in the United States consist of fewer than 50 cows (McBride and Greene, 2009), some larger operations housing up to 6,000 milking cows exist. Due to the need for managing large groups of cows without access to reproductive hormones and estrous synchronization protocols, these herds face unique issues in their breeding programs. Moreover, the prohibited use of antimicrobial therapies for reproductive tract infections represents another management limitation.

Characterizing the incidence of RD in a population of organic certified Holstein cows housed in large herds would provide the opportunity for an adequate comparison with large conventional herds, as size and breed would be similar. In addition, specific management factors resulting from organic regulations, affecting reproductive health and performance could be identified.

Therefore, the objective of this study was to determine the lactational incidence risks of RD including dystocia (**DYS**), retained fetal membranes (**RFM**), puerperal metritis (**PM**), clinical endometritis (**CE**), pyometra (**PYO**), late pregnancy loss (**LPL**), and culling due to reproductive reasons (**CRR**) in large ORG with predominant use of Holstein genetics. In addition, potential risk factors for RD, including parity number, previous milk yield (**PMY**), season of dry-off, access to grazing at dry-off, length of the dry period (**DPL**), length of the close-up period (**CUPL**), calculated gestation length (**CGL**), season of calving, access to grazing at calving, concurrent health conditions, and total number of artificial inseminations (**nAI**) were analyzed.

MATERIALS AND METHODS

Study Population and Herd Management

Data from cows in 8 USDA organic certified dairy herds under the same overall management and supervision were included in this study. Retrospective information from completed lactations starting with calvings occurring between January 1, 2016, and December 31, 2018, was retrieved from the on-farm recording system software (PCDART, Dairy Records Management Systems, Raleigh, NC). Herds were in northern Colorado (n = 6) and western Texas (n = 2), and Holstein was the predominant breed (>97%). During the study period, the herd size on the farms ranged between 900 and 6,000 cows. General demographics of the study herds including herd size, parity distribution, and average milk yield are presented in Table 1.

Cows were milked 2 to 3 times/d, in rotary, parallel, or herringbone milking parlors. A TMR was fed twice daily to meet or exceed the nutritional requirements for lactating Holstein cows producing 30 kg/d of milk (3.5% fat and 3.1% true protein; NRC, 2001). The grazing season in the herds located in Colorado extended from May 3 to September 23, from April 26 to October 8, and from April 23 to October 5 in 2016, 2017, and 2018, respectively. In the Texas herds, the grazing season extended from May 26 to September 15, from April 6 to September 23, and from April 5 to September 27 in 2016, 2017, and 2018, respectively. The diets used in the participating dairies varied through seasons and locations. However, some components were included in the general formulation of the rations across farms. As a reference, the prepartum diets were based on corn silage (10.5–16%, DM basis), wheat silage (10.5–16.6%), grass hay (14.2–20%) alfalfa hay (32–41%), grain mix (12.6–19.2%), and mineral and vitamin mix (3.8%). Anionic salts were included into the ration (dietary

cation anion difference = −100 mEq/kg). Nongrazing diets for lactating cows were based on corn silage (14–17.5%); wheat silage (13–20%); grain mix containing soybean, soy hulls, corn wheat, mineral, and vitamins (47.5–50.5%); sorghum silage (3.0–4.5%); alfalfa hay (12–16%); and grass hay (0–3%). During grazing, the diets included corn silage (5–7%); wheat silage (17–19%); grain mix containing soybean, soy hulls, corn, wheat, and minerals and vitamins (38–41%); sorghum silage (5 to 7%); alfalfa hay (2%); grass hay (0–1.5%); and pasture grazing (estimated as 33–40% of DMI). Although the composition and pasture management differed among the participating dairies, in general, the grazing management included rotational grazing on perennial alfalfa, annual ryegrass, Italian ryegrass, teff grass, oat ryegrass, sorghum pasture, and native grass pasture.

All herds had similar health-monitoring programs, where trained farm personnel supervised by the attending veterinarian performed diagnosis, treatment, and recording of health conditions. Standardized disease definitions documented on a common standard operating procedures document including consistent times of assessment were used across herds. Therapies were consistent with the US organic regulations.

No artificial hormones were used to manage reproduction, and cows were bred based on AI using visual estrous detection. After 45 DIM, all the cows were moved into insemination groups. To assess estrous behavior, cows' tailheads were painted daily with colored chalk and checked for chalk removal, which indicated mounting behavior. If estrus was determined, cows were inseminated during the morning. Cows with >4 breedings or diagnosed open at 180 DIM were moved to pens with bulls for natural service. If cows in the bull pens were diagnosed pregnant, they were moved to pens housing only pregnant cows. The voluntary waiting period was set to 45 DIM. Depending on herd location, pregnancy diagnosis was performed by the attending veterinarian by transrectal palpation of the uterus and its contents at 60 d after AI or on a weekly basis for cows in the bull pens. In some locations, an early pregnancy diagnosis was performed at 32 d after AI by transrectal ultrasound examination. Pregnancy was rechecked by transrectal palpation before dry-off.

Data Editing

Finalized lactations were considered the experimental units. Data sets were organized into records with the information starting at the dry-off date from the previous lactation in multiparous cows (**MP**) or at calving in primiparous cows (**PP**). Lactations were considered completed if they had a subsequent dry-off, death, or culling date reported. For the calculation of lactational incidence risk for RD, the periods at risk were determined considering published literature from conventional herds. Events reported during time periods in conflict with the disease definitions were removed due to potential misclassification bias (Dohoo et al., 2009). Finally, lactations missing calving date or with conflicting calving and dry-off dates (e.g., previous dry-off date occurring after calving date), were excluded from the data set.

Outcomes Variables and Case Definitions

Reproductive disorders including DYS, RFM, PM, CE, PYO, and LPL, as well as CRR were the main outcome variables considered in the analyses. Disease events were diagnosed and recorded by trained animal health care personnel. The attending veterinarians developed disease definitions, protocols, recording codes, and provided regular training to health care workers.

The diagnosis criteria met the generally accepted definitions for these RD and were uniform across herds. Calving difficulty was recorded in a 5-point scale: 1 = spontaneous or normal, 2 = slight problem or no major intervention, 3 = needed assistance, 4 = considerable effort, or 5 = extremely difficult. For analysis purposes,

Table 1. Description of the data set considered in the retrospective analyses of herds at organic certified dairy farms

Herd ID[1]	Location	Available records (n)	Primiparous cows (n, %)	Multiparous cows (n, %)	Rolling herd milk average (kg)
1	CO	20,398	8,083 (39.6)	12,315 (60.4)	8,909
2	CO	8,664	3,880 (44.8)	4,784 (55.2)	
3	CO	2,659	208 (7.8)	2,451 (92.2)	
4	CO	2,002	124 (6.2)	1,878 (93.8)	
5	CO	1,822	7 (0.4)	1,815 (99.6)	
6	CO	4,207	1,347 (32)	2,860 (68)	7,996
7 and 8	TX	24,916	8,229 (33)	16,687 (67)	9,310

[1]Herds 1 to 5 had the same overall supervision but were individually managed. Herds 3 to 5 were intended for housing multiparous cows. Herd 6 had independent supervision and management. Herds 7 and 8 maintained combined records from 2 geographically contiguous dairies located in western Texas.

a binary classification of DYS (dystocia = calving score ≥3) was created based on Mee (2008) and Vieira-Neto et al. (2017). Retained fetal membranes were diagnosed when cows failed to release the fetal membranes within 24 h after calving. Puerperal metritis was defined as a cow with fetid watery red-brown uterine discharge and systemically ill with pyrexia and anorexia within the first 21 DIM (Sheldon et al., 2006). Clinical endometritis was defined as purulent vaginal discharge characterized by more than 50% of pus flecks in cows not systemically ill at ≥21 d after calving (Sheldon et al., 2006). Pyometra was defined as the presence of purulent fluid in the uterus, in the presence of a corpus luteum and a closed cervix (Sheldon et al., 2006). As diagnosis of pregnancy by transrectal ultrasound at 30 to 35 d after AI (with reconfirmation at 60 d after AI) was not consistently used across the study herds, early pregnancy loss was not considered in the analyses. Consequently, LPL was defined as a nonpregnant diagnosis in cows that were previously diagnosed pregnant. Most cases were determined during re-checking of cows returning to heat behavior or at the time of dry-off (Keshavarzi et al., 2017). Finally, CRR was defined as cows dying due to reproductive disease or as cows leaving the herd due to uterine problems or failing to conceive after 300 DIM, with concurrent low milk production (<13.6 kg/d).

Explanatory Variables

Potential risk factors for RD included in the analyses were parity number, 305-d mature-equivalent (305-d ME) milk yield during the previous lactation (**PMY**; kg), season of dry-off, access to grazing at dry-off, DPL, CUPL, CGL, season of calving, access to grazing at calving, concurrent health conditions, and total nAI (categorized as 1, 2 or 3, and >3).

Categories of PMY were created based on 305-d ME records as low (<mean − SD), medium (mean ± SD), and high (>mean + SD), based on milk yield averages from each study herd. Season of dry-off and calving were categorized as occurring in spring, summer, fall, or winter. Specific grazing dates from the participating herds were applied to create a binary category indicating if dry-off or calving occurred during grazing (generally extended between April and October).

Length of the dry period was categorized as short (<30 d); intermediate (30–70 d); or long (>70 d), following ranges in previous reports (Mayasari et al., 2017; O'Hara et al., 2019). The CUPL was categorized as short (<7 d), intermediate (7–14 d) and long (>14 d). The CGL was defined as the difference between the date of calving and the most recent previous AI or, in the case of natural service, the estimated recorded breeding. Based on the distribution of CGL in the study population and according to published ranges (Norman et al., 2011), CGL was categorized as short (<270 d), intermediate (270–285 d), and long (>285 d).

Information about health conditions concurrent with RD was retrieved from farm records and only health events occurring simultaneously or before the diagnosis of RD were considered in the analyses. Concurrent conditions included ketosis (hyperketonemia symptoms or high levels of BHB, as indicated by handheld strips; Overton et al., 2017), digestive problems (acidosis, bloating, bloody manure, constipation and scours, displaced abomasum), lameness, respiratory problems, and other RD. Number of AI was categorized as 1, 2 or 3, and >3. The distributions of finalized lactations by category of risk factors are presented in Table 2.

Statistical Analyses

All the statistical procedures were performed in SAS 9.4 (SAS institute Inc., Cary, NC). Descriptive analyses considered lactational incidence risk and time-to-event analyses. Because the time at risk of all the outcomes

Table 2. Frequency distribution of the available lactation records, divided by explanatory variables considered as potential risk factors for reproductive disorders in the study farms

Variable	Categories	Frequency (%)
Parity	Primiparous	33.8
	Multiparous	66.2
Previous lactation milk yield[1]	Low	15.5
	Medium	68.8
	High	15.7
Season of dry-off	Spring	25.5
	Summer	30.0
	Fall	23.4
	Winter	21.1
Grazing at dry-off	No	51.2
	Yes	48.9
Length of the dry period	<30 d	36.2
	30–70 d	47.8
	>70 d	16.0
Close-up period length	<7 d	19.9
	7–14 d	25.6
	>14 d	54.6
Calculated gestation length	<270 d	8.0
	270–285 d	69.5
	>285 d	22.4
Season of calving	Spring	22.8
	Summer	27.9
	Fall	26.6
	Winter	21.7
Grazing at calving	No	53.3
	Yes	46.7
Number of AI	1	35.2
	2–3	37.7
	>3	27.1

[1]Categorization based on 305-d mature-equivalent milk averages in each study herd; low (<mean – SD), medium (mean ± SD), and high (>mean + SD).

was shorter than the observation period (from calving to dry-off), incidence risk was an appropriate measure of disease frequency (Dohoo et al., 2009). Univariate calculations of incidence risk were used to describe overall disease frequency (PROC FREQ). Additionally, univariate survival analysis (PROC LIFETEST) was used to determine the Kaplan-Meier median DIM to the first occurrence of each reproductive event per lactation.

Associations between RD and the proposed explanatory variables were assessed through multivariable logistic regression. We used the generalized estimated equation approach through the GENMOD procedure of SAS. As the observations (lactations) were clustered in herds where cows had common environment and management, standard errors were corrected using herd as repeated statement. The exchangeable correlation structure was included in the procedure, which assumes the correlation to be the same among observations within a cluster (herd; Dohoo et al., 2009). Additionally, a binomial distribution, logit link, and covariate patterns for model fit were requested in the statistical procedure.

The analyses considered parity category and variables related to health, season of calving, grazing effects, and reproductive events during the lactation as explanatory variables for RD. However, due to the lack of precalving information (dry period and PMY) in PP cows, 2 separate groups of models were fitted. The first group of models included the whole study population and was used to investigate the effect of parity, season of calving, grazing at calving, nAI, and concurrent health conditions as follows:

$$\text{Logit}\ [\pi\ (\text{Y}_{\text{RD}} = 1)] = \beta_0 + \beta_1(\text{Parity}) + \beta_2(\text{Season of calving}) + \beta_3(\text{Grazing at calving}) + \beta_4\ (\text{nAI}) + \beta_{x1,\ldots,\ xn}(\text{Concurrent health conditions}),$$

where Y_{RD} is the response variable (occurrence of each RD considered in the study) and β_0 is the intercept of the model. The reference categories for the explanatory variables (β_1– β_3) were MP cows, winter season, and nongrazing calving, >3 inseminations, and healthy for specific diseases, respectively.

A second group of models investigating the effects of dry period management and PMY were fitted for MP cows as follows:

$$\text{Logit}\ [\pi\ (\text{Y}_{\text{RD}} = 1)] = \beta_0 + \beta_1(\text{PMY}) + \beta_2(\text{Season of dry-off}) + \beta_3(\text{Grazing at dry-off}) + \beta_4(\text{DPL}) + \beta_5(\text{CUPL}) + \beta_6(\text{CGL}),$$

where Y_{RD} is the response variable (occurrence each RD considered in the study) and β_0 is the intercept of the model. The reference categories for the explanatory variables (β_1–β_5) were high milk yield, winter season, and nongrazing at dry-off, >70 d, >14 d, and 275 to 285 d, respectively.

For each RD, a maximum model with all possible predictors was identified. The final model was selected through backward elimination. Significant predictors were selected at *P*-value <0.05 and controlling variables remained in the model at *P*-value ≤0.20.

RESULTS

Lactational Incidence Risk and Time of Disease Diagnosis

After edits, 64,668 out of 66,311 lactation records were available for the analyses. The total number of individual cows in the data set was 40,920, where 55.6, 31.2, and 13.2% of the study cows had 1, 2, and 3 lactations available, respectively. The number of cases and the lactational incidence risks for specific RD and CRR are presented in Table 3. Clinical endometritis had the greatest lactational incidence risk (10.1%), followed by DYS (8.8%), PM (4.9%), PYO (4.0%), LPL (3.6%), and RFM (2.0%). Lactational incidence risk of CRR was 7.5% considering all the lactations at risk, whereas CRR represented 29.1% of the overall culling.

The median and the first and third quartile values for time to diagnosis of RD are presented in Table 4. As DYS and RFM occur during parturition and within 24 h postpartum, respectively (Gilbert, 2016), survival analyses for these RD were not performed. Consequently, the earliest RD diagnosed in lactation was PM, with 50% of the cases reported at 6 DIM. Median time to diagnosis for CE was 23 DIM. Pyometra cases occurred later in the lactation, with median time to diagnosis of 111 DIM, whereas median time to diagnosis for LPL was 228 DIM. Culling due to low reproductive performance occurred with a median of 391 DIM.

Risk Factors for Reproductive Disorders

Significant predictors of RD, as well as controlling variables included in the logistic regression models, are presented in Tables 5 and 6. The odds of DYS, PM, and CE were greater in primiparous cows compared with multiparous cows [OR = 1.13 (95% CI = 1.01–1.3), 1.4 (1.1–1.8), and 1.2 (1.1–1.4), respectively].

Previous milk yield category was associated with DYS, CE, and CRR. Smaller odds of DYS and CE were observed in the low [DYS = 0.80 (0.6–0.9); CE = 0.74 (0.63–0.9)] and medium [DYS = 0.8 (0.7–0.9);

Table 3. Number of cases and lactational incidence risk (95% CI) of reproductive disorders among multiparous and primiparous cows observed in cows calving between January 2016 and December 2018 in a total of 64,668 lactations from 8 organic certified dairy herds

	Number of cases		Lactational incidence risk (%, 95% CI)	
Reproductive disorder	Multiparous	Primiparous	Multiparous	Primiparous
Dystocia[1]	1,996	782	8.5 (8.2–8.9)	9.5 (8.9–10.1)
Retained fetal membranes	943	332	2.2 (2.1–2.3)	1.5 (1.4–1.7)
Puerperal metritis	1,609	1,541	3.8 (3.6–3.9)	7.0 (6.7–7.2)
Clinical endometritis	3,933	2,563	9.2 (8.9–9.5)	11.7 (11.3–12.1)
Pyometra	1,818	745	4.3 (4.1–4.4)	3.4 (3.2–3.7)
Late pregnancy loss[2]	1,637	646	3.8 (3.7–4.0)	3.0 (2.8–3.2)
CRR[3]	3,313	1,508	7.7 (7.5–8.0)	6.9 (6.6–7.2)
CRR[4]	3,313	1,508	27.0 (26.2–27.8)	35.2 (33.8–36.6)

[1]Dystocia scores were available for 31,330 lactations.

[2]The analyses for pregnancy loss included 64,122 lactations.

[3]Culling due to reproductive causes; incidence risk calculated considering all lactations at risk.

[4]Culling due to reproductive causes; incidence risk calculated among culled cows.

CE = 0.8 (0.7–0.93)] PMY categories compared with high-producing cows during their previous lactation. Conversely, cows in the low and the medium PMY categories had 1.6 (1.4–1.8) and 1.14 (1.02–1.3) times greater odds of CRR compared with high PMY cows (Tables 5 and 6).

Season of dry-off was associated with the occurrence of RFM, PM, CE, PYO, and LPL. The odds of RFM for winter were the greatest compared with the other 3 seasons (Table 5). On the contrary, for PM spring and fall dry-off had 1.2 (1.16–1.3) and 1.7 (1.1–1.9) times greater odds of PM compared with winter, respectively. No differences were found between summer and winter. For CE, differences were only observed for fall versus winter, where fall dry-off had 0.48 (0.4–0.6) times lower odds of CE (Table 6) than dry-off in winter. Greater odds of PYO were observed in spring [1.5 (1.2–1.8)] and summer [1.6 (1.4–1.8)] dry-off compared with dry-off in winter (Table 6). Finally, in LPL spring dry-off had lower odds of pregnancy loss compared with winter dry-off [0.7 (0.6–0.8)], whereas no differences were observed among other seasons (Table 6).

Table 4. Summary of Kaplan-Meier survival analyses for time to diagnosis (DIM) for reproductive disorders that occurred during January 2016 to December 2018 in eight large organic certified dairy herds

	Percentile		
Reproductive disorder (DIM)	25	50	75
Puerperal metritis	4	6	8
Clinical endometritis	21	23	25
Pyometra	101	111	156
Late pregnancy loss	158	228	297
CRR[1]	323	391	433

[1]CRR = culling due to reproductive causes.

The effect of access to grazing at dry-off was associated with PM, PYO, and LPL. Cows that were not exposed to grazing at dry-off had greater odds of being diagnosed with these disorders (Table 5 and 6). Length of the dry period was associated with all RD, except for PYO and CE. For DYS, the 30 to 70 d DPL had 1.3 (1.2–1.4) times greater odds of DYS compared with a DPL exceeding 70 d. For RFM, PM, and LPL a shorter DPL was found to be a risk factor compared with the reference category (DPL > 70 d, Tables 5 and 6). Finally, a DPL ranging between 30 and 70 d resulted in a protective effect for CRR compared with a longer DPL (Table 6).

Time in the close-up groups was associated with all RD considered in this study. Shortened CUPL (<7 d) was associated with greater odds of DYS [1.4 (1.1–1.8)], RFM [2.1 (1.8–2.4)], PM [1.9 (1.6–2.1)], PYO [1.5 (1.2–1.8)], LPL [2 (1.7–2.3)], and CRR [1.4 (1.04–1.8)] compared with CUPL exceeding 14 d (Tables 5 and 6). For CE, a CUPL of 7 to 14 d had greater odds of disease compared with CUPL > 14 d.

Calculated gestation length was associated with presentation of DYS, RFM, PM, CE, and CRR. Shortened and longer CGL resulted in greater odds of DYS and PM compared with the middle CGL range (Table 5). On the other hand, the odds of RFM were multiplied by 3.2 (2.4–4.1) in cows with CGL lower than 270 d compared with the middle CGL category. Longer CGL increased the odds of CE and CRR by 1.5 (1.1–2.0) and by 1.2 (1.1–1.4) times, respectively.

Season of calving was associated with the occurrence of PM, PYO, and LPL. Calving in summer resulted in greater odds of PM (1.13, 1.1–1.2) and PYO (1.7, 1.12–2.5) compared with winter calving, whereas fall calving had smaller odds of LPL compared with winter 0.80 (0.60–0.96; Tables 5 and 6).

Cows with access to pasture at calving had greater odds of RFM [1.3 (1.2–1.5)], PM [1.1 (1.01–1.2)], and CRR [1.2 (1.17–1.3)], compared with cows calving in the nongrazing season (Tables 5 and 6). Conversely, access to grazing at calving had a protective effect for LPL, where cows calving during grazing season had lower odds of LPL [0.5 (0.4–0.3); Table 6].

The smallest category for nAI (nAI = 1) was associated with smaller odds of PYO [0.5 (0.40–0.60)], LPL [0.26 (0.20–0.30)], and CRR [0.9 (0.89–0.91)] compared with cows with ≥3 (Table 6). Cows in the mid nAI (2 or 3) category also had lower odds of LPL [0.80 (0.70–0.99)] and CRR [0.63 (0.60–0.70)] relative to greater category of nAI.

Concurrent health problems had a significant effect in most RD. Dystocia was strongly associated with RFM. Cows with DYS had 4.6 (3.0–6.9) times greater odds of RFM compared with cows with normal calving. The odds of PM were greater in cows diagnosed with RFM [7.7 (5.4–11)], ketosis [1.6 (1.3–2.1)], and digestive disorders [1.2 (1.1–1.6)], compared with unaffected cows. Significant diseases associated with the occurrence of CE were DYS [1.6 (1.5–1.6)], RFM [1.2 (1.1–1.3)], PM [2.8 (1.9–3)], and ketosis [1.2 (1.1–1.5)]. Presentation of RFM was associated with LPL [2.6 (1.7–4)], whereas PM [1.9 (1.3–2.7)], and CE [6.0 (3.9–9.1)] were associated with PYO. Cows diagnosed with PM and PYO had greater odds of CRR [1.3 (1.2–1.4) and 2.8 (2.2–3.5), respectively].

DISCUSSION

Postpartum diseases commonly occur in dairy cows and their detrimental effect on welfare, fertility, and survival are widely recognized (Stojkov et al., 2015; Gilbert 2016). However, as dairies under organic cer-

Table 5. Associated risk factors for dystocia, retained fetal membranes, and puerperal metritis[1]

	Dystocia			Retained fetal membranes			Puerperal metritis		
Variable	OR[2]	95% CI	*P*-value	OR	95% CI	*P*-value	OR	95% CI	*P*-value
Parity									
Primiparous	1.13	1.01–1.30	0.03				1.40	1.10- 1.80	0.007
Multiparous	—	—	—	—	—	—	—	—	—
Previous milk yield									
Low	0.80	0.60- 0.90	0.0009						
Medium	0.80	0.70- 0.90	0.008						
High	—	—	—	—	—	—	—	—	—
Season of dry-off									
Spring				0.80	0.70–0.96	0.02	1.20	1.16–1.30	<0.0001
Summer				0.50	0.40–0.70	<0.0001	0.90	0.70–1.10	0.3
Fall				0.50	0.40–0.60	<0.0001	1.70	1.10–1.90	<0.0001
Winter	—	—	—	—	—	—	—	—	—
Grazing at dry-off									
No	1.23	0.90–1.70	0.2				1.20	1.04–1.30	0.008
Yes	—	—	—	—	—	—	—	—	—
Length of the dry period									
<30 d	1.10	0.80–1.50	0.4	2.10	1.60–2.70	<0.0001	1.40	1.20–1.70	0.0008
30–70 d	1.30	1.20–1.40	<0.0001	1.10	0.80–1.40	0.6	1.13	1.01–1.30	0.03
>70 d	—	—	—	—	—	—	—	—	—
Close-up period length									
<7 d	1.40	1.10–1.80	0.01	2.10	1.80–2.40	<0.0001	1.90	1.60–2.10	<0.0001
7–14 d	1.03	0.95–1.13	0.45	1.20	1.10–1.40	<0.001	1.20	1.00–1.50	0.07
>14 d	—	—	—	—	—	—	—	—	—
Calculated gestation length									
<270 d	1.50	1.30—1.80	<0.0001	3.20	2.40–4.10	<0.0001	2.40	1.70–3.30	<0.0001
270–285 d	—	—	—	—	—	—	—	—	—
>285 d	1.50	1.40–1.70	<0.0001	0.90	0.60–1.30	0.7	1.50	1.20–2.00	0.002
Season of calving									
Spring							1.10	1.00–1.12	0.13
Summer							1.13	1.10–1.20	<0.0001
Fall							1.03	1.00–1.10	0.11
Winter	—	—	—	—	—	—	—	—	—
Grazing at calving									
No	—	—	—	—	—	—	—	—	—
Yes	0.90	0.70–1.00	0.12	1.30	1.20–1.50	<0.0001	1.10	1.01–1.20	0.03

[1]All models included dairy as clustering factor; cells with a dash represent the reference category; risk factors with empty cells were not significantly associated with the reproductive disorder in the column head; controlling variables ($P \leq 0.2$) are included in the table and in the models; risk factors from concurrent health conditions are reported in the manuscript.

[2]OR = adjusted odds ratios.

Table 6. Associated risk factors for clinical endometritis, pregnancy loss, pyometra, and culling due to low reproductive performance (CRR)[1]

	Clinical endometritis			Pyometra			Late pregnancy loss			CRR[2]		
Variable	OR	95% CI	*P*-value	OR	95% CI	*P*-value	OR	95% CI	*P*-value	OR	95% CI	*P*-value
Parity												
Primiparous	1.20	1.10–1.40	0.003									
Multiparous	—	—	—	—	—	—	—	—	—	—	—	—
Previous milk yield												
Low	0.74	0.63–0.90	0.0008							1.60	1.40–1.80	<0.0001
Medium	0.80	0.70–0.93	0.005							1.14	1.02–1.30	0.02
High	—	—	—	—	—	—	—	—	—	—	—	—
Season of dry-off												
Spring	0.80	0.50–1.20	0.28	1.50	1.20–1.80	0.0003	0.70	0.60–0.80	<0.0001			
Summer	0.84	0.50–1.40	0.5	1.60	1.40–1.80	<0.0001	0.77	0.50–1.20	0.2			
Fall	0.48	0.40–0.60	<0.0001	1.00	0.80–1.30	0.99	0.90	0.70–1.10	0.23			
Winter	—	—	—	—	—	—	—	—	—	—	—	—
Grazing at dry-off												
No				1.10	1.03–1.30	0.01	1.70	1.40–2.10	<0.0001	1.80	1.50–2.10	<0.0001
Yes	—	—	—	—	—	—	—	—	—	—	—	—
Length of the dry period												
<30 d	1.30	1.00–1.70	0.05				2.50	1.50–4.20	<0.001	1.30	0.90–1.80	0.16
30–70 d	1.10	1.00–1.20	0.05				1.20	0.96–1.40	0.14	0.74	0.60—0.90	<0.0001
>70 d	—	—	—	—	—	—	—	—	—	—	—	—
Close-up period length												
<7 d	1.50	1.00–2.30	0.5	1.50	1.20–1.80	0.0002	2.00	1.70–2.30	<0.0001	1.40	1.04–1.80	0.02
7–14 d	1.10	1.01–1.14	0.01	1.30	1.20–1.50	<0.0001	1.20	0.90–1.60	0.18	1.04	0.96–1.10	0.3
>14 d	—	—	—	—	—	—	—	—	—	—	—	—
Calculated gestation length												
<270 d	1.03	0.90–1.20	0.7							0.97	0.84–1.10	0.6
270–285 d	—	—	—	—	—	—	—	—	—	—	—	—
>285 d	1.50	1.10–2.00	0.02							1.20	1.10–1.40	0.006
Season of calving												
Spring				1.10	0.70–1.70	0.6	1.03	0.90–1.13	0.6			
Summer				1.70	1.12–2.50	0.01	1.13	0.93–1.40	0.21			
Fall				1.20	0.80–1.70	0.4	0.80	0.60–0.96	0.02			
Winter	—	—	—	—	—	—	—	—	—	—	—	—
Grazing at calving												
No	—	—	—	—	—	—	—	—	—	—	—	—
Yes				1.40	0.90–2.10	0.14	0.50	0.40–0.70	<0.001	1.20	1.17–1.30	<0.0001
Number of AI at conception												
1				0.50	0.40–0.60	<0.0001	0.26	0.2–0.30	<0.0001	0.90	0.89–0.91	<0.0001
2–3				1.00	0.80–1.20	0.78	0.80	0.7–0.99	0.04	0.63	0.60–0.70	<0.0001
>3	—	—	—	—	—	—	—	—	—	—	—	—

[1]All models included dairy as clustering factor; cells with a dash represent the reference category; risk factors with empty cells were not significantly associated with the reproductive disorder in the column head; controlling variables ($P \leq 0.2$) are included in the table and in the models; risk factors from concurrent health conditions are reported in the manuscript; OR = adjusted odds ratio.

[2]CRR = culling due to reproductive causes.

tification have regulations leading to unique management practices, it is not clear whether the information that originated from conventional herds may be fully applicable to organic systems. In fact, available studies comparing reproductive efficiency between conventional and ORG are not conclusive, and most reports include moderate-size populations restricted to specific geographical areas (Pinedo and Velez, 2019). Consequently, the aims of this study were to determine lactational incidence risks of RD in a group of US large ORG, as well as to identify potential risk factors that can be targeted as management interventions for improving productive efficiency and animal wellbeing.

The lactational incidence risk of DYS in our study (8.8%) was similar to that reported by Ribeiro et al. (2013) in a prospective study performed in conventional herds (8.5%). On the contrary, Gröhn and Rajala-Schultz (2000) reported a lower lactational incidence risk of DYS (2.1%), which may be explained by the retrospective nature of that study. As expected, in the current analysis, PP cows had greater odds of DYS and high-producing cows were also at greater risk of DYS, which agrees with data presented by Gröhn and Rajala-Schultz (2000) and Daros et al. (2017).

Dystocia has been associated with poor mineral, energetic, and immune profiles, which are factors that also increase the risks of other disorders, such as RFM, PM, and culling in dairy cattle (Gröhn and Rajala-Schultz, 2000; Gilbert, 2016; Pinedo et al., 2017). In the present study, we observed that shorter time in the close-up groups was a risk factor for DYS. Therefore, an inadequate duration of the pre-calving conditioning period may affect energetic status, body condition, and consumption of pre-calving diets, which are related to postpartum clinical and subclinical diseases (Roche et al., 2009; Ribeiro et al., 2013; Chebel et al., 2018).

Another important factor associated with DYS was CGL. The associations presented here agree with studies performed in different populations where short and long CGL are associated with abnormal calving, stillbirths, and insufficient productive performance (Norman et al., 2009; Nogalski and Piwczynski, 2012). Due to the potential effect of extended GL on fetus-pelvic disproportion, GL is determinant factor of DYS in herds where Holstein is the predominant breed, as it is the case of the study population used in this study (Gröhn and Rajala-Schultz, 2000; Mee, 2008). More relevant, the use of natural service and the prohibition on the use of estrous synchronization drugs in ORG poses a challenge for efficiently predicting calving dates, and therefore, for planning dry-off and close-up management (Pinedo and Velez, 2019). Consequently, record management become a key strategy for peripartum management and disease prevention. In the case of this population, the effect of imprecisely estimated conception dates, due to the use of natural service, is more relevant that the actual length of gestation. Mistaken due dates originated from these inaccuracies, would result in shortened or extended dry-off and close-up periods, with detrimental effects on the transition period.

Previous studies have reported incidences of RFM ranging from 4 to 15% (Gröhn and Rajala-Schultz, 2000; Hooshmandabbasi et al., 2018) and 18.5 to 27.6% for PM (Santos et al., 2015), which is higher than the incidence observed in this study (Table 3). In this population, lower RFM incidences were determined for winter dry-off, whereas shortened DPL, CUPL, and CGL were associated with increased RFM, which agrees with findings associated with DYS.

The study herds recorded PM cases when cows had abnormal watery-fetid vaginal discharge and systemic illness, usually fever (Sheldon et al., 2006). In similar studies with seasonal grazing herds, the incidence of PM ranged between 11 to 29.7%, and the associated risk factors were parity category, DYS, and RFM (Giuliodori et al., 2013; Daros et al., 2017). In this study, the lactational incidence risk of PM was 4.9%, likely due to the definition restricted to systemically compromised cows. Risk factors for PM were RFM, season of dry-off, shortened close-up time, and DPL, and CGL, which had not been previously reported in organic herds.

In the participating herds, CE was defined according to the vaginal discharge score described by Sheldon et al. (2006) and by Giuliodori et al. (2013a). In the current study, an incidence of 10.1% was determined, which is within the reported range of 5.8 and 33% in conventional dairy herds with herd sizes between 300 and 1,000 lactating cows (Dubuc et al., 2010; Giuliodori et al., 2013a).

The main reported risk factors for the development of CE are DYS, NEB, metritis, and other postpartum diseases (Dubuc et al., 2010; Cheong et al., 2011; Giuliodori et al., 2013a). We observed that ketosis and PM were risk factors for CE, which agrees with the results presented by Cheong et al. (2011) and Gilbert (2016). Cheong et al. (2011) determined that cows affected by ketosis had increased odds of subclinical endometritis, indicating that ketosis may alter the endometrium homeostasis and making cows susceptible to infections.

Pyometra represents about 2 to 5% of the uterine disease in dairy cows (Karstrup et al., 2017) and the lactational incidence is about 4% (Gilbert, 2016), which is similar to the value reported in this study (Table 3). Pyometra is particularly relevant to ORG because hormonal and antimicrobial treatments are not available (El-Tahawy and Fahmy, 2011; NOP, 2019), resulting in an increased risk for reproductive culling in affected cows. Although PYO tended to occur in mid and late

lactation in this population (Table 4), there were significant associations with season of dry-off and calving, grazing at dry-off and at calving, DPL, and CUPL. Table 6 shows all *P*-values for these significant associations. However, these associations are questionable because of the extended time between the exposure factor and the outcome. Regarding PYO, concurrent health conditions such as PM increased the odds of PYO by 1.9, whereas CE increased the odds by 6 times. Both PM and CE are recognized as risk factors of PYO (Sheldon et al., 2006; Karstrup et al., 2017).

Pregnancy loss is significant in dairy cows and the reported prevalences vary from 6 to 39%, with most losses occurring during the first trimester of gestation (Chebel et al., 2004; Moore et al., 2005; Hernandez et al., 2012). In our study, we observed a lactational risk of 3.6% for LPL. Naturally, the low occurrence reported in the current study could be explained by the exclusion of early embryo mortality, which represent most of pregnancy loss in dairy cattle (Wiltbank et al., 2016).

Parity number, stage of lactation, calving month, pregnancy stage, and previous LPL have been recognized as risk factors for pregnancy loss (Keshavarzi et al., 2017). Norman et al. (2012) analyzed almost 8.5 million lactations from US herds, and reported that LPL occurs more frequently before 175 DIM, which is only in partial agreement with the timing observed in this study, where 50% of the cases occurred before 228 DIM. In addition to potential reproductive health underlying conditions, the association between greater nAI and increased odds of LPL suggest a link with the use of natural service, as nonpregnant cows are moved to bull-groups, which may relate with infection with pathogens that cause LPL (Campero et al., 2003; Klauck et al., 2016).

Failure to conceive or remain pregnant is a major reason why cows are culled (Bascom and Young, 1998; Hadley et al., 2006). In a study analyzing data from cows in 2,054 US herds, the annualized live culling rate and death rate were 25.1 and 6.6%, respectively (Pinedo et al., 2010). In that population, reproduction represented 17.7% of the overall culling, which is smaller than the value determined in our study (29% of the total culling). This greater representation within the overall culling could be explained by the prohibition of antibiotic and hormonal therapies for RD that would result in infertility and subsequent departure from the herd.

As for most of the RD evaluated here, DPL and the amount of time spent under close-up management were significant risk factors for CRR. These factors present a particular challenge for large ORG, as the combination of AI at detected estrus and the use of natural service affects the ability to predict the subsequent calving date, making it difficult to schedule dry-off and close-up movements (Pinedo and Velez, 2019).

Organic regulation mandates access to grazing during a minimum of 120 d/yr (NOP, 2019). During this period, large ORG adjust their management to provide adequate water access and facilitate rumen adaptation to low nonstructural carbohydrates and high fermentable NDF diets (Roche et al., 2011; McDougall et al., 2018). Other challenges include the potential for increased heat stress and the need for performing health and reproductive monitoring in restricted time windows (Pinedo and Velez, 2019). These conditions may affect cows' performance, as access to pasture modifies their locomotion behavior, with increased standing and walking times (Stafford and Gregory, 2008; Legrand et al., 2009). Moreover, the change from nongrazing diets to grazing involves complex metabolic adaptations with potential consequences for health (Grala et al., 2019).

Considering that the transition period is a critical time for cow health and survival, we tested the effect of occurrence of calving during the grazing season on subsequent reproductive outcomes. As anticipated, cows that calved in that period had greater odds of occurrence of some RD (Tables 5 and 6).

In the same idea, some authors suggest that the ideal BCS in grazing cows should be about 3.0 to 3.5 to minimize health conditions such as metritis and to improve general performance (Ribeiro et al., 2013; Daros et al., 2017). These standards are relevant for ORG because, in general terms, cows with calving during the grazing season had greater odds of PM and PYO (Tables 5 and 6), suggesting that these animals may face additional challenges during the transition period, which may predispose them to deeper negative energy balance.

CONCLUSIONS

Lactational incidence risks of reproductive diseases were within published ranges in conventional herds. In this study, the main risk factors associated with reproductive diseases were length of the dry period, length of the close-up period, calculated gestation length, and concurrent health problems. Additionally, an effect of season and access to grazing at calving on the occurrence of some RD was observed. The identified risk factors open opportunities for interventions, especially focused in precalving management, estimation of due dates, and closer monitoring of pregnancies from natural services in large organic dairy herds.

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